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Managed Pressure Cementing of 11-3/4" Liner in a Narrow Margin Well: Overcoming Losses and Isolation Challenges

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Abstract

This paper presents the successful application of Managed Pressure Cementing (MPC) and formation-strengthening fluids to achieve zonal isolation in a development well in TAL Block, North Pakistan. Due to severe Tectonic stresses in this region, weak formation and very narrow pore frac pressure window the Liner RIH itself was a challenge and job criticality was increased. The objective was to safely cement an 11-3/4" liner in a depleted, loss-prone formation where traditional cementing practices posed high risks of failure and formation breakdown.

The cementing strategy combined optimized fluid sequencing and MPC pressure control to address ongoing losses and narrow annular clearance. The open hole was drilled using low-density OBM, followed by a carefully engineered fluid train comprising a specialized spacer to act as hydrostatic buffer, loss cure spacer to prevent the possibility of losses, a lightweight lead slurry, and a dense tail cement. Only four centralizers were deployed due to annular restrictions (0.29"), with flow path further restricted by the liner hanger. Real-time ECD simulation guided pump rates and displacement volumes, while MPC equipment was used to maintain bottomhole pressure within the safe margin across known weak zones.

The operation was executed despite several significant challenges including severe losses during drilling, a narrow 0.29" annulus, limited centralization, and reduced TFA due to the liner hanger assembly. By managing the ECD through MPC and fluid hierarchy, cement was successfully placed above the liner top without inducing losses. No backflow was observed after displacement, and cement returns were consistent with expected volumes. Post-job evaluations confirmed cement presence at the designed depth, with the top of liner tagged successfully. A formation integrity test (FIT) conducted at 2.19 SG EMW demonstrated that the formation could now withstand much higher pressures than previous failed attempts at 1.74 SG. The liner cementing was declared a success, eliminating the need for remedial work and enabling the resumption of the drilling program. This job validated the strategic use of MPC and specialty fluids in overcoming narrow-margin and loss-prone conditions.

This case marks one of the first documented uses of MPC combined with loss cure fluid technology in this field. It demonstrates how integrating real-time downhole pressure control with adaptive fluid design can overcome extreme operational constraints. The paper offers a valuable reference for executing complex

liner cementing jobs in depleted or narrow-margin wells where conventional cementing methods may not be viable.

Introduction

Zonal isolation during well construction is essential for controlling fluid migration, ensuring casing support, and protecting sensitive formations. Achieving reliable isolation becomes particularly challenging in wells with depleted zones, narrow annular clearances, and high fluid loss tendencies. In such environments, conventional cementing methods often fall short, requiring more adaptive and pressure-managed approaches.

The Case presented precisely this scenario during its 11-3/4" liner cementing phase. After drilling through highly unstable formations with known loss zones most notably at depths of 1,900 m, 1,915 m, and 2,250 m, the cementing design had to be re-engineered. Total losses had been encountered below the 13-3/8" casing shoe and repeat attempts at plugging had limited success. As a result, standard hydrostatically driven cementing posed an unacceptably high risk of inducing losses or failing to isolate the liner shoe.

To mitigate these risks, the team implemented MPC technique. MPC is increasingly used in wells with narrow margins between pore pressure and fracture gradient, particularly in mature or fractured reservoirs. This method allows real-time control of annular pressure, enabling full circulation and controlled ECD during cementing, especially useful in liner cementing operations where fluid density and displacement pressure must be tightly managed (Siddiqi et al. 2016; Awang et al. 2022).

In the MKD 03 case, the cementing design incorporated a lightweight 10.84 ppg lead slurry and a high-density 15.0 ppg tail slurry, separated by dual unweighted spacers tailored for loss mitigation and fluid compatibility. The job was executed using an MPD system to control surface backpressure dynamically and keep downhole ECD within a safe range.

Previous field cases, such as those reported by (Awang et al. 2022), have shown that MPC can enable the successful placement of long casing strings in highly depleted offshore wells, while maintaining circulation and ensuring adequate bonding. Similarly, (Siddiqi et al. (2016) demonstrated MPC's effectiveness in onshore Middle East environments, where cementing success required careful control of circulating pressures and fluid sequencing. These case studies, alongside deepwater Gulf of Mexico examples (JPT 2025), reinforce MPC's growing value as a cementing technique in complex wells.

This paper documents the complete case history of the MKD 03 11-3/4" liner cementing job, from design rationale to field execution, and provides practical insights for applying MPC in similar high-risk environments.

Well Challenges

Following the cementing of the 13-3/8" casing, it was anticipated that the Murree Formation would end and a deeper formation (Lockhart) capable of supporting higher mud weights for setting the 9-5/8" casing shoe. However, TAL Block is characterized by mountainous terrain and highly variable subsurface profiles. The Murree Formation persisted beyond the expected depth and comprises red-variegated clays and medium-grained, ferruginous sandstone with interbedded claystone and siltstone streaks, reflecting an arid, terrestrial depositional environment. To address the extended presence of this formation and maintain well integrity, an 11-3/4" liner was planned to run in hole, bridging the gap between the 13-3/8" casing and the planned 9-5/8" casing shoe.

Losses were encountered during drilling at multiple intervals: just below the previous casing shoe, around 1,900–1,915 m, and severely at 2,250 m. These loss zones were partially treated with LCM pills, thixotropic and conventional cement plugs. 11-3/4" liner cementing job was executed under various risks that made zonal isolation challenging. These risks originated from both subsurface uncertainties and surface limitations, many of which directly affected design assumptions and operational flexibility.

Formation Challenges

Severe and Persistent Loss Zones. Drilling losses were first observed immediately below the 13-3/8" casing shoe, and further aggravated at depths of 1,900 m, 1,915 m, and severely at 2,250 m. Cement plugs were carried out to cure some of these loss zones, but drilling continued with repeated losses up till the TD at 2,553 m. It was diagnosed there is no specific zone in the open hole suspected of the losses after marking the TD so it was assumed that the losses are present in all of the open hole.

Narrow Pore-Frac Gradient Window. Simulations during planning phase of the cement job confirmed a very tight mud weight window, with pore pressure equivalent to 10.42 ppg and formation breakdown (loss inducing) pressures slightly above 11.17 ppg. The operating window (0.75 ppg) was too narrow to allow for conventional cement job hydrostatic pressures, making the job highly sensitive to equivalent circulating density (ECD) during the cement job.

Wellbore Geometry Complexity. The open hole section was drilled with 12.2" bit and under-reamed to a mean diameter of 14.88" (as recorded by the caliper log) with localized enlargements up to 15.4". These washout zones pose a significant threat to zonal isolation by affecting the displacement efficiency of the cement job. The challenge was compounded by the presence of the exceedingly low stand-off and rate limitations due to the employment of the MPC further increased the risk of incomplete mud removal.

Mechanical and Operational Limitations

Annular Clearance and Centralization Constraints. The liner's OD (11.75") and the casing's ID (12.347") allowed for just 0.298" clearance. This severely restricted the placement of centralizers. As a result, the liner was RIH with only 2 centralizers at the shoe track, leading to a very low standoff being observed across the remaining length of the liner. The liner also could neither be rotated nor reciprocated due to frictional pressure concerns. The low total flow area across the Liner hanger also increased the ECDs at the top of the liner.

Managed Pressure Cementing monitoring. MPD system was installed on the rig to maintain annular pressures and avoid losses. During the job, backpressure between was maintained in such a way that the ECD remains below 11.17 ppg at the weakest point i.e 2,246 m. This required precise coordination between rig, cementing crew, and MPD team throughout the cement job.

Job Design

The MPC job was designed keeping in view the density and rheology hierarchy of the cementing fluid train. The displacement rate was calculated using the software in such a way that the ECDs at the weakest point do not cross 11.17 ppg while also applying surface back pressure (SBP). The cementing operation was designed with an unweighted spacer of 8.34 ppg, serving as a pre-flush and reducing the hydrostatic head in the annulus. This was followed by a 10.42 ppg loss-circulation spacer to mitigate potential losses. The cement placement sequence comprised a 10.84 ppg lead slurry, designed to minimize ECD in the open hole, followed by a 15.0 ppg tail slurry to provide high compressive strength at the shoe.

Advanced cementing simulator was used to check the SBP proposed by the MPD team and the displacement rates were adjusted to keep the equivalent pressure of 11.17 ppg across the weakest zone 2,246 m. All the fluids were planned to be pumped at 4 BPM except the displacing fluids. During the final displacement sequence, 385 bbl of 10.42 ppg mud was initially pumped at a rate of 4 bpm to displace the cement slurry. This was followed by 30 bbl of 10.42 ppg spacer at 3 bpm to maintain fluid separation at the TOL and prevent contamination. Displacement then continued with an additional 54.4 bbl of mud at a reduced rate of 2.5 bpm while also applying the SBP. For the final stage, 20 bbl of mud was pumped at 2

bpm to gently bump the liner wiper plug on the landing collar, ensuring cement placement without inducing any influx or losses.

Cement volumes were calculated using the hole geometry recorded in the Caliper log and adjusted for maximum realistic standoff. Due to restricted annular clearance, approximately 0.298 inches per side in the overlap zone, centralizers deployment was restricted.

Job Simulation

The Job simulations were performed using cementing software to validate pressure response, placement behavior, and displacement volumes. Real-time job support included the use of cement unit calibration, density verification, and interface tracking. MPC modeling ensured pressure compliance under varying flow rates and fluid weights.

Rate vs ECDs at different loss zones.

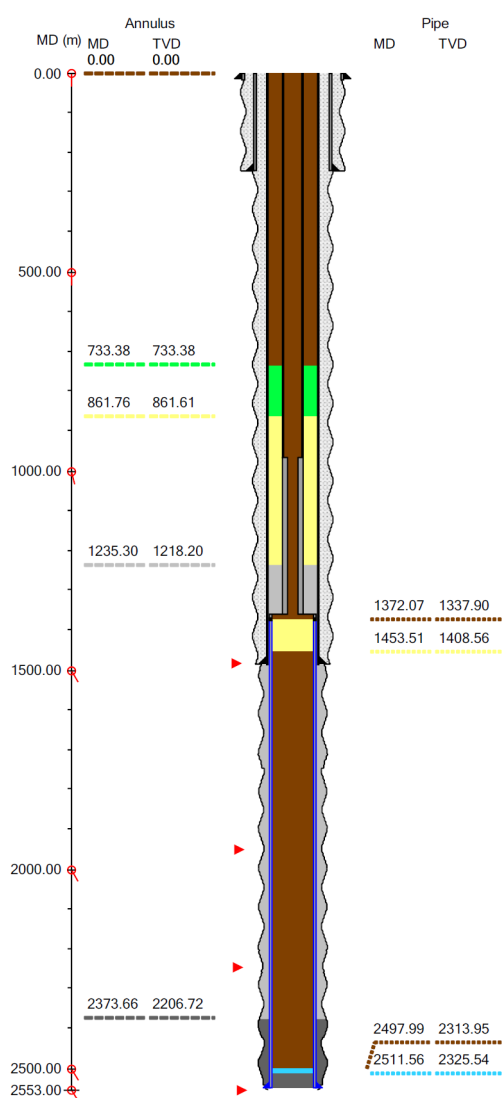


Figure 1—Wellbore Schematics after the cement job

Table 1—Rates vs ECDs

Vol. (bbl)	Q - in (bpm)	Q - out (bpm)	Pump P. (psi)	ECD @ 1483.00 m (S.G.)	ECD @ 1950.00 m (S.G.)	ECD @ 2246.00 m (S.G.)	ECD @ 2553.00 m (S.G.)	P. @ 2553.00 m (psi)
0.732	4.00	4.00	326	1.316	1.307	1.304	1.301	4372
23.435	4.00	4.00	429	1.316	1.307	1.304	1.301	4372
46.138	4.00	4.00	533	1.316	1.307	1.304	1.301	4372
68.309	4.00	4.00	570	1.316	1.307	1.304	1.301	4372
91.012	4.00	4.00	548	1.316	1.307	1.304	1.301	4372
113.715	4.00	4.00	461	1.316	1.307	1.304	1.301	4372
136.418	4.00	4.00	366	1.316	1.307	1.304	1.301	4372
159.121	4.00	4.00	366	1.316	1.307	1.304	1.301	4372
181.823	4.00	4.00	366	1.316	1.307	1.304	1.301	4372
204.394	4.00	4.00	396	1.316	1.307	1.304	1.301	4372
227.097	4.00	4.00	381	1.316	1.307	1.304	1.301	4372
249.800	4.00	4.00	365	1.316	1.307	1.304	1.301	4372
272.503	4.00	4.00	357	1.316	1.307	1.304	1.301	4372
295.206	4.00	4.00	362	1.316	1.307	1.304	1.301	4372
317.909	4.00	4.00	358	1.316	1.307	1.304	1.301	4372
340.612	4.00	4.00	354	1.316	1.307	1.304	1.301	4372
363.315	4.00	4.00	350	1.316	1.307	1.304	1.301	4372
386.018	4.00	4.00	346	1.316	1.307	1.304	1.301	4372
408.720	4.00	4.00	342	1.316	1.307	1.304	1.301	4372
431.423	4.00	4.00	338	1.316	1.307	1.304	1.301	4372
454.126	4.00	4.00	334	1.316	1.307	1.304	1.301	4372
476.829	4.00	4.00	330	1.316	1.307	1.304	1.301	4372
498.859	4.00	4.00	283	1.316	1.307	1.304	1.301	4372
521.562	4.00	4.78	91	1.324	1.314	1.310	1.300	4371
544.265	4.00	5.51	91	1.355	1.338	1.331	1.308	4396
566.788	4.00	4.45	54	1.321	1.311	1.307	1.286	4323
589.491	4.00	3.04	54	1.306	1.299	1.291	1.277	4293
612.194	4.00	2.97	54	1.305	1.299	1.283	1.276	4287
634.601	4.00	4.00	144	1.316	1.307	1.284	1.282	4309
657.304	4.00	4.00	125	1.316	1.307	1.282	1.281	4306
680.007	4.00	4.00	128	1.316	1.301	1.283	1.282	4308
702.710	4.00	4.00	127	1.316	1.290	1.283	1.282	4308
725.413	4.00	4.00	133	1.316	1.283	1.282	1.282	4311
748.116	4.00	4.00	145	1.316	1.283	1.282	1.285	4319
770.819	4.00	4.00	158	1.316	1.284	1.283	1.288	4328
793.522	4.00	4.00	138	1.300	1.273	1.274	1.280	4304
816.225	4.00	4.00	117	1.272	1.261	1.265	1.273	4279
838.928	4.00	4.00	138	1.264	1.265	1.271	1.278	4296
861.630	4.00	4.00	186	1.281	1.279	1.286	1.291	4340
884.333	4.00	4.00	196	1.281	1.281	1.288	1.293	4346
907.036	4.00	4.00	206	1.280	1.283	1.290	1.295	4352
929.739	4.00	4.00	216	1.280	1.286	1.292	1.297	4358
951.788	4.00	4.00	196	1.279	1.288	1.294	1.298	4363
973.248	2.50	2.50	146	1.271	1.283	1.289	1.293	4347
987.438	2.50	2.50	129	1.268	1.280	1.286	1.291	4338
1001.627	2.50	2.50	352	1.370	1.361	1.357	1.355	4555
1015.816	2.50	2.50	409	1.371	1.361	1.357	1.365	4588
1029.019	2.00	2.00	403	1.351	1.345	1.343	1.361	4575
1040.370	2.00	2.00	451	1.352	1.346	1.343	1.369	4602
1046.455	2.00	2.00	478	1.352	1.346	1.343	1.373	4616

Laboratory testing

Cementing fluids were designed at 150 deg F as proposed by the software by incorporating the temperature recorded in the caliper log after the shut in time period of 14 hrs. The lead and tail cement slurries were formulated to keep by keeping low fluid loss < 50 ml/30 min. and cement strength more than 1,100 psi for the lead cement slurry and 1,800 psi for the tail cement slurry.

Thickening Time.

Table 2—Slurry Thickening Time assumptions

Slurry type	Batch (hr:min)	Mixing Time	Placement Time	Set packer, POOH and reverse out excess cement	Safety Margin	Min TT @ 30 Bc	Actual TT @ 30 Bc
Lead Slurry		00:00	03:52	01:30	X1.5	08:03	08:58
Tail Slurry		0:30	02:39	-	+2:00	05:09	05:30

Job execution

Design vs Actual.

Table 3—Design vs Actual

Details	Planned	Actual
Unweighted CemBOND Spacer		
Density (ppg)	8.42	8.42
Rate (bpm)	4.0	~4.0
Volume (bbl)	50	50
CemSHIELD Spacer		
Density (ppg)	10.42	10.42
Rate (bpm)	4.0	~4.0
Volume (bbl)	150	150
LW Lead Slurry		
Density (ppg)	10.84	10.84
Rate (bpm)	4.00	~4.00
Volume (bbl)	293	293
Top Of Cement (m)	1,248	Cement contamination found during the Bit RIH
Conventional Tail Slurry		
Density (ppg)	15.0	15.0
Rate (bpm)	4.00	~4.00
Volume (bbl)	60	60
Top Of Cement (m)	2,373	N/A
Displacement Fluids		
Density (ppg)/	8.34 (Water) / 12.8 (Mud) / 10.84 (Spacer) / 12.8 (Mud)	8.34(Water) / 12.8(Mud) / 10.84 (Spacer) / 12.8 (Mud)
Vol. (bbl)	5/385/30/74.4	5/385/30/80
Rate (bpm)	4/3/2.5/2	~4/3/2.5/2
Final Pressure (psi)	490	530
Returns		
Mud / Cement / Spacer (bbl)	Contaminated	Pure Mud
Post Job		
Plug Bumped	Yes	No (upon tag it was found ~30 m shallower)
Volume of returns (bbl)	0	No Losses observed

The maximum pore pressure was equal to 10.42 ppg and minimum frac was 11.17 ppg. Just before the cementing job the well was hydrostatically underbalanced and SBP was applied to maintain the required pressure downhole without inducing losses. Once the heavier cementing fluids start to enter into the annulus, the SBP decreased gradually to accommodate for the increased frictional pressure and the heavy weight of the spacer and the cement. The SBP was maximum when the rate was slowed down to observe the pin shearing when the liner Dart latches onto the liner wiper plug and then start to wipe the liner displacement. The SBP was also increased towards the end of the job when the rate is slowed to down to observe the plug bump.

Job Graph

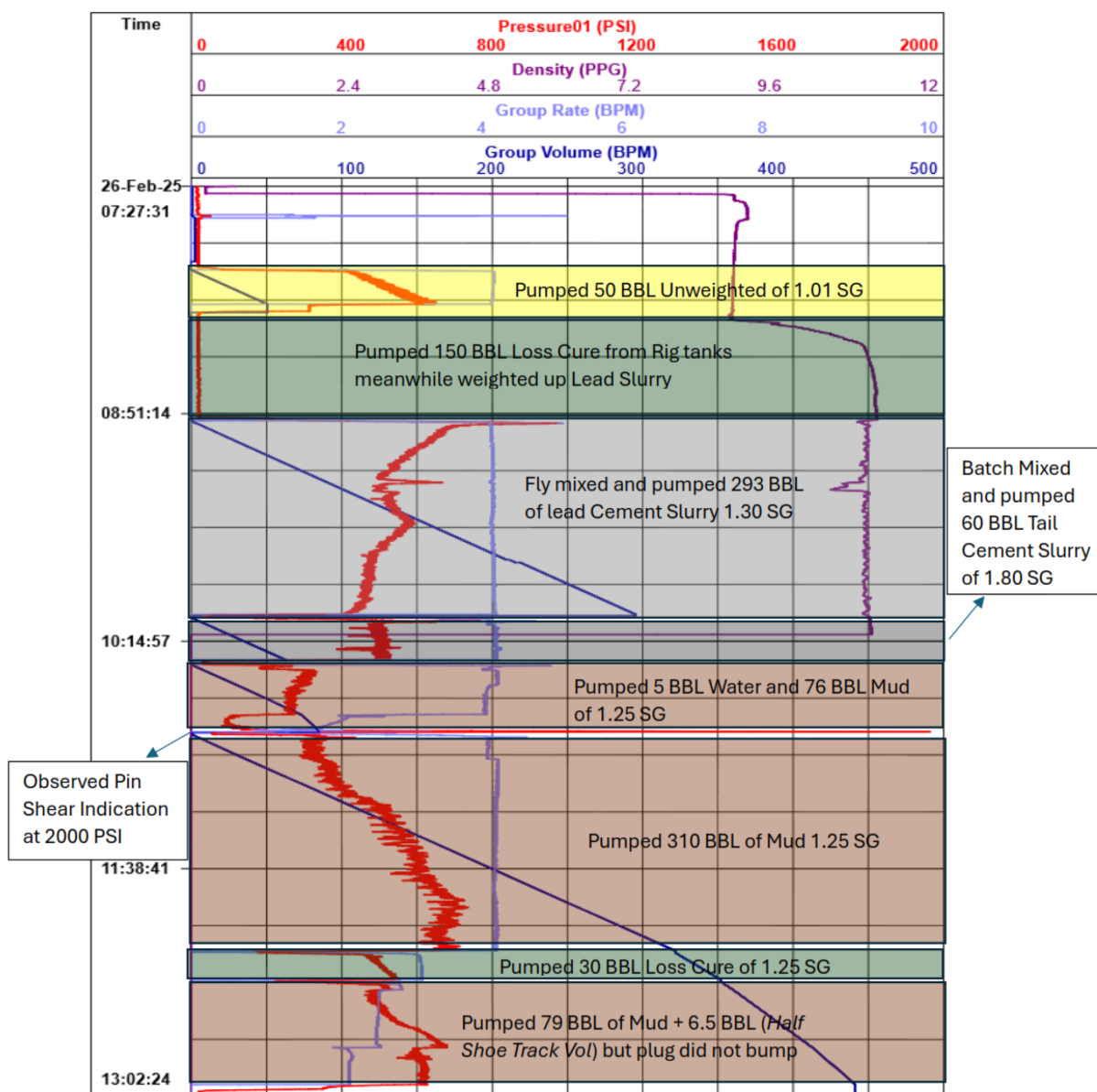


Figure 2—Cement Job Graph

Conclusion

The 11-3/4" liner cementing job at MKD-03 was executed successfully despite challenging formation conditions and operational constraints. Cement was placed across the liner shoe without inducing losses, and

the well remained stable throughout the job. Zonal isolation was confirmed at the shoe where the centralizers were placed, cement rise above the overlap section, and no formation losses occurred during the operation. The combined use of lightweight slurry, stage-wise spacers, and MPC enabled effective placement in a high-risk environment. Operational discipline, pressure management, and fit-for-purpose fluid design contributed to a technically successful and safe outcome.

Zonal isolation at the shoe was confirmed through a Shoe Bond Test (SBT) using 18.34 ppg EMW. No backflow or communication was observed, indicating a solid cement seal across the critical interval. This confirmed that the primary objective of achieving the shoe integrity.

Cement was found at top of the liner during bit RIH, which aligned with the planned top of cement inside the 13-3/8" casing. This confirmed cement presence in the liner overlap and verified that returns were effectively displaced.

The Managed Pressure Cementing (MPC) system was key to this outcome. ECD was maintained below 11.17 ppg, and surface backpressure was adjusted in real time to match downhole conditions. The MPC system allowed full circulation with no indication of loss or gain.

The dart was launched and tracked successfully. While no plug bump was observed, pressure response and volume tracking indicated shear pin activation within the expected range.

The cementing job was completed in 5 hours and 13 minutes without losses, safety incidents, or non-productive time. All parameters remained within plan, and operational coordination between teams was effective.

Overall, the cementing operation achieved its key goals. The shoe was isolated, cement reached the overlap as planned, and the well remained stable throughout. The design, execution, and pressure control all contributed to a clean and technically sound outcome.

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